

# Mohamed Alaa M. Ahmed

✉ mohame43@purdue.edu 📞 +201508253312 📍 Cairo, Egypt 🌐 LinkedIn 🏠 GitHub

## RESEARCH INTERESTS

Trustworthy AI, Machine Learning Security, Large Language Models, Human-AI Interaction, and Edge AI Systems.

## EDUCATION

### Purdue University, Bachelor of Computer Engineering

Sep 2021 – May 2026 | IN, USA

- GPA: 4.0/4.0; Dean's List & Semester Honors — Fall 2024, Spring 2025
- Graduate-Level Coursework: Reinforcement Learning, Machine Learning, Data Mining, Applied Quantum Computing, Introduction to Communication Networks (*In Progress*), Artificial Intelligence (*Planned Spring 2026*)

### Egypt University of Informatics, Bachelor of Computer Engineering

Sep 2021 – Jul 2026 | Cairo, Egypt

- GPA: 3.9/4.0; Inaugural cohort, selected among ~30 students nationwide through a presidential merit-based scholarship awarded to the top high-school performers out of ~650,000 students across Egypt.
- Earned dual-degree year at Purdue University (Aug 2024 – May 2025) sponsored by Microsoft Egypt.

## RESEARCH EXPERIENCE

### Undergraduate Researcher,

Aug 2024 – May 2025 | IN, USA

Purdue VIP, Duality Lab (Advisor: Prof. James Davis)

- Led a four-member research team investigating adversarial robustness in LLMs, designing and implementing novel hybrid attack pipelines by integrating Greedy Coordinate Gradient (GCG) with Prompt Automatic Iterative Refinement (PAIR) to exploit alignment vulnerabilities.
- Achieved up to 91.6% Attack Success Rate (ASR) on Llama-3 (a 33.2% increase over PAIR's baseline), with hybrid methods consistently matching or exceeding the performance of state-of-the-art single-mode attacks.
- Demonstrated that these hybrid strategies reliably pierced advanced SOTA defenses such as Gradient Cuff and JBSHield, which fully blocked constituent single-mode attacks (Token-level and Prompt-level).
- Benchmarked 4 open-source LLMs across 3 attack/defense methodologies using SorryBench dataset, and reported critical findings that exposed vulnerabilities transferability in current safety stacks to OpenAI and Meta Bug Bounty Programs, contributing to responsible vulnerability disclosure and mitigation.
- Work presented in the poster track at IEEE SecDev 2025 (Indianapolis, USA).

### Research Assistant for Human-AI Interaction,

Dec 2024 – May 2025 | IN, USA

Purdue University, College of Liberal Arts (Advisor: Prof. Hyesun Choung)

- Architected and implemented the ICC Tutor, a conversational AI system leveraging Retrieval-Augmented Generation (RAG), Few-Shot Persona Adaptation, and a dual-module framework to deliver psychologically safe and personalized intercultural communication training.
- Deployed the system on Azure using OpenAI, LangChain, Svelte (JavaScript) and TypeScript; tested with 25 students in a university communication course and received highly positive feedback.
- Key Findings: Qualitative analysis (using an LLM-as-a-Judge coding framework) affirmed that the non-judgmental AI agent fostered high psychological safety and engagement, with differentiated benefits observed across novice and expert user groups.
- Research paper submitted to ACM IUI 2026 (first author; *under review*); poster accepted at AAAI 2026.

## PROFESSIONAL EXPERIENCE

### Software Engineering Intern – AI Multi-Agent Systems, Technowelle GmbH

Jul 2025 – Present | Cairo, Egypt

- Built a GraphRAG system (FAISS + LangChain + VDB) to manage 100+ embedded system requirements, increasing retrieval precision & response consistency ~30% in multi-agent workflows.
- Developed a multilingual on-device voice assistant for smart device control using NER + fine-tuned LLMs for intent classification and context-aware dialogue.
- Fine-tuned & quantized LLMs for edge deployment, shrinking model size and latency by 85%+ with minimal accuracy loss.
- Integrated Model Context Protocols (MCP) for cooperative reasoning across embedded testing agents, reducing latency and improving scheduling efficiency.
- Conducted research on prompt engineering & context management for distributed agent systems, supporting low-latency trustworthy edge AI.

### Machine Learning Engineer Intern - Computer Vision, Electro Pi

Mar 2024 – Aug 2024 | Cairo, Egypt

- Implemented a high-accuracy Arabic OCR system for digitizing ancient manuscripts, combining CNN-Transformer architectures with character-level language modeling to handle complex cursive and degraded scripts, achieving a 60% improvement in text recognition accuracy over baseline Tesseract models.
- Built large-scale action recognition pipelines using MMAAction2 and YOLOv8, incorporating temporal convolutional networks (TCNs) and 3D CNNs for multi-frame motion understanding in human activity monitoring and industrial collision detection.

TEACHING EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Undergraduate Teaching Assistant</b> , <i>ECE 20875: Python for Data Science</i> ,<br><i>Purdue University (Instructor: Prof. Qiang Qiu)</i> <ul style="list-style-type: none"><li>Supported labs and office hours for ~300 students/semester.</li><li>Guided assignments/projects in Python, regular expressions, NumPy, and pandas.</li></ul>                                                                                                  | Aug 2024 – May 2025 |
| <b>Machine Learning Instructor</b> , <i>Electro Pi</i> <ul style="list-style-type: none"><li>Delivered weekly sessions on advanced computer vision models (R-CNN, YOLO, and Florence) and multimodal vision-language systems; guided students in building and deploying interactive ML applications using Streamlit.</li><li>Provided consultation hours to support end-to-end model development, optimization, and deployment workflows.</li></ul> | Mar 2024 – Aug 2024 |
| <b>Junior Teaching Assistant</b> , <i>ECE 115: Digital Systems Design</i> ,<br><i>Egypt University of Informatics</i> , (Instructor: Prof. Amr ElSherif) <ul style="list-style-type: none"><li>Supported GTAs in labs and held office hours for ~90 students.</li><li>Mentored first-year engineers on logic circuit design and debugging circuit logical errors.</li></ul>                                                                         | Sep 2023 – Jan 2024 |

LEADERSHIP & SERVICE

|                                                                                                                                                                                                                                                                                                                            |                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>AI Committe President, Vice Chairman</b> , <i>IEEE Computer Society EUI SBC</i><br>Organized the IEEE Data Science Bootcamp 500+ registrants and coordinated a Kaggle competition with 30 teams; established a professional roadmap and delivered hybrid sessions guiding students toward careers in the data industry. | Sep 2023 – Present |
| <b>Co-Founder</b> , <i>IEEE EUI Student Branch</i><br>Grew a community of 300+ students, organizing AI-focused workshops, hackathons, and speaker sessions, while coordinating major IEEE events including JAC-ECC 2023, DSC MENA 2024 and NASYP Region 8.                                                                 | Jun 2023 – Present |

PUBLICATIONS

|                                                                                                                                                                                                                                                                                                             |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Advancing Jailbreak Strategies: A Hybrid Approach to Exploiting LLM Vulnerabilities and Bypassing Modern Defense</b>  ,<br><i>IEEE Secure Development Conference (SecDev 2025) – Poster Track, Indianapolis, USA.</i> |  |
| <b>Adaptive AI for Personalized Intercultural Communication Education: A Conversational Agent Powered by Retrieval-Augmented Generation</b> , <i>AAAI 2026, Singapore.</i><br>EGSAI Workshop – Student Abstract and Poster Program                                                                          |  |

HONORS & AWARDS

|                                                                                                                                                                                                                                                                                          |             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>AAAI-2026</b> <ul style="list-style-type: none"><li><i>Student Scholar and Volunteer Scholarship and Travel Grant</i></li><li><i>EGSAI Workshop 2026:</i><ul style="list-style-type: none"><li><b>Best Student Poster Award</b></li><li><b>3MT Contest Winner</b></li></ul></li></ul> | Jan 2026    |
| <b>Full Microsoft Merit Scholarship (EUI–Purdue Dual Degree)</b> ,<br><i>Egypt University of Informatics &amp; Microsoft: (1-year) Tuition &amp; living costs.</i>                                                                                                                       | Apr 2024    |
| <b>Presidential Excellence Scholarship – Egypt University of Informatics</b> , (5 years) <i>Full coverage of tuition and transportation, including three blended courses with Purdue Online, awarded for Egypt’s Top High School Graduates.</i>                                          | Sep 2021    |
| <b>Purdue OUR Scholarship</b> ,<br><i>Office of Undergraduate Research &amp; College of Libral Arts</i>                                                                                                                                                                                  | Dec 2024    |
| <b>Dean’s List &amp; Semester Honors – Fall 2024, Spring 2025</b> , <i>Purdue University</i>                                                                                                                                                                                             | 2024 – 2025 |
| <b>EUI Top Achiever and President List</b> , <i>Egypt University of Informatics</i>                                                                                                                                                                                                      | Oct 2023    |
| <b>ECPC Finalist</b> , <i>ICPC (Egyptian Collegiate Programming Contest)</i>                                                                                                                                                                                                             | Aug 2023    |

SKILLS

|                                                                                                                                     |  |
|-------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Programming Languages:</b> Python, C, C++, CUDA, Rust, SQL, Java, JavaScript, TypeScript                                         |  |
| <b>Data Analysis &amp; Visualization:</b> Pandas, NumPy, Matplotlib, Seaborn, Plotly, Tableau                                       |  |
| <b>Machine Learning:</b> Logistic Regression, KNN, SVM, Random Forest, XGBoost (Scikit-learn)                                       |  |
| <b>Deep Learning &amp; Computer Vision:</b> CNN, LSTM, R-CNN, ResNet, YOLO, OpenCV, TensorFlow, PyTorch                             |  |
| <b>Large Language Models (LLMs):</b> Hugging Face, LangChain, (RAG), Few-shot & (CoT) Prompting, Fine-tuning, Quantization          |  |
| <b>Cloud &amp; Deployment:</b> Microsoft Azure, AWS, GCP, Docker, FastAPI, FastMCP, Streamlit, Git, GitHub Actions, CI/CD pipelines |  |

CERTIFICATES

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Microsoft Azure AI Fundamentals  , Machine Learning specialization, DeepLearning.AI  , Applied Data Science Specialization, IBM  , AWS Cloud Foundations  , Applied Data Science Lab, WorldQuant University  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|